

DBMS Project Report

Indian Agriculture & Crop Production Database

Student: Dev Manjibhai Sheta

Roll No.: 23BCP110

Program: B.Tech Computer Science & Engineering

Pandit Deendayal Energy University (PDEU), Gandhinagar

Team Size (N): 1 | Database Engine: MySQL 8.0

A. Introduction

This project implements a relational database for storing and analyzing India's district-level agricultural crop production data. The system captures area cultivated, production quantity, and yield for crops grown across every state and district of India, segmented by season and year, over a 27-year period (1997-2023).

Project Scope

The scope covers conceptual design (ER modeling), logical design (relational schema with normalization), physical implementation in MySQL, an automated data-loading pipeline, and a graded set of analytical SQL queries. Physical-level tuning (indexing, query optimization) and database security/backup automation were intentionally scoped out to keep the project focused on core relational design and querying, which are this lab's primary learning objectives.

Objectives

- Design a normalized (3NF) relational schema for real-world agricultural data
- Correctly model the fact table conceptually as a ternary relationship between District, Crop, and Season
- Build an automated ETL pipeline to clean and load a large public dataset into MySQL
- Develop a graded range of SQL queries — from basic retrieval to subquery-based analysis
- Demonstrate sound database design principles: normalization, referential integrity, and data quality validation

Tools and Technologies Used

Component	Choice
RDBMS	MySQL 8.0 (via MySQL Workbench)
ETL / Automation	Python 3 (pandas, mysql-connector-python)
Data Source	data.gov.in / data.desagri.gov.in (Govt. of India open data)
Development OS	macOS
Documentation	Markdown, Word (this report)

Unlike a synthetic or toy dataset, this project uses real government data, which surfaced a genuine data-quality problem (detailed in Section g) that had to be identified and resolved — giving the project practical, not just theoretical, database engineering value.

B. Requirements Analysis

Data Requirements — what the database stores

- States and their districts (740 districts across 34 states/UTs)
- Crops grown, with their type classification (Cereal, Pulses, Oilseeds, etc.)
- Cropping seasons (Kharif, Rabi, Whole Year, Summer, Autumn, Winter)
- Yearly production records: area cultivated, total production, and yield, per district/crop/season/year combination

Functional Requirements — what users query the database for

- Aggregate production statistics by state, district, crop, or season
- Trends over time (year-over-year production/yield changes)
- Comparative analysis (which states/districts are most productive, most diverse)
- Outlier and threshold-based lookups (highest production records, above-average yields)
- Insertion of new yearly records without duplicating existing ones

Non-Functional Requirements

- Referential integrity must be enforced between every production record and its state/district/crop/season
- The schema must scale to the full 27-year, 740-district dataset (~455,000+ records) without redesign
- Data loading must handle missing/malformed values gracefully rather than failing on the first bad row

Use Case Summary

Actor	Use Case
Agricultural analyst	Query production trends by state/crop/year
Policy researcher	Compare crop diversity and yield across regions
Database administrator	Load new yearly data; maintain referential integrity

C. Conceptual Design — ER Diagram

The diagram below uses standard Chen ER notation: rectangles for entities, ellipses for attributes (primary keys underlined), and a diamond for the relationship, with cardinalities and participation constraints marked on each connecting line.

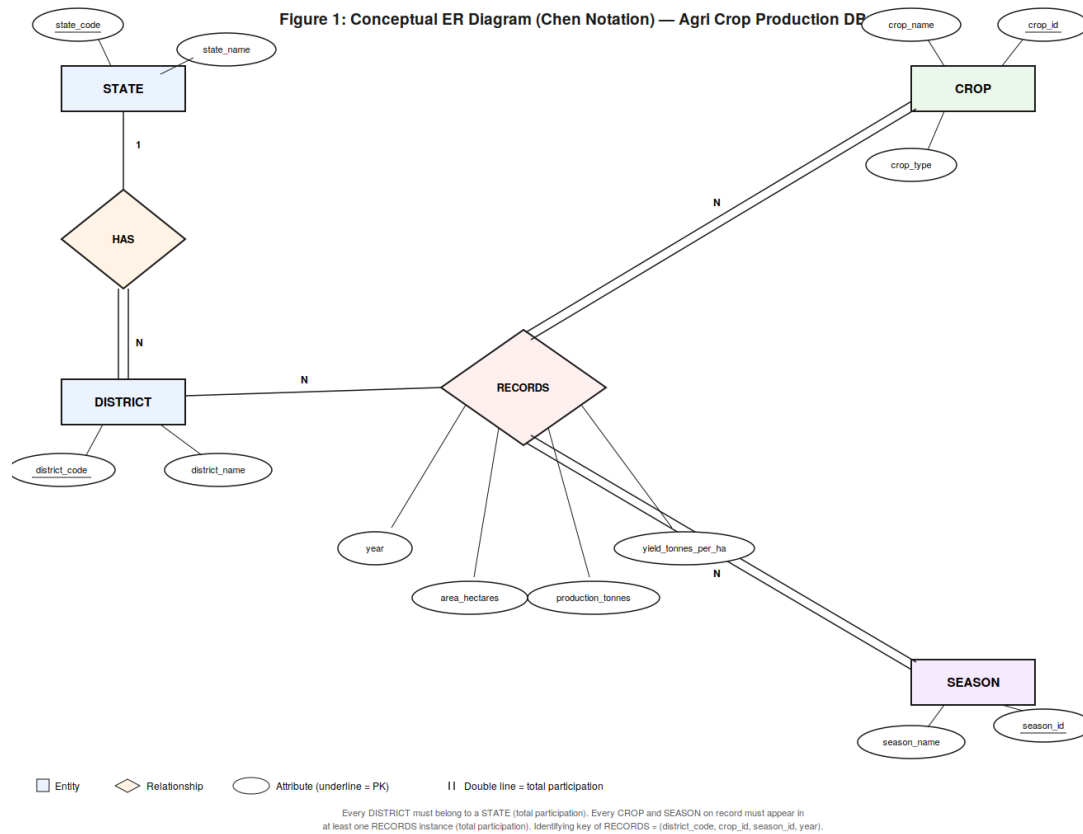


Figure 1: Conceptual ER Diagram (Chen Notation)

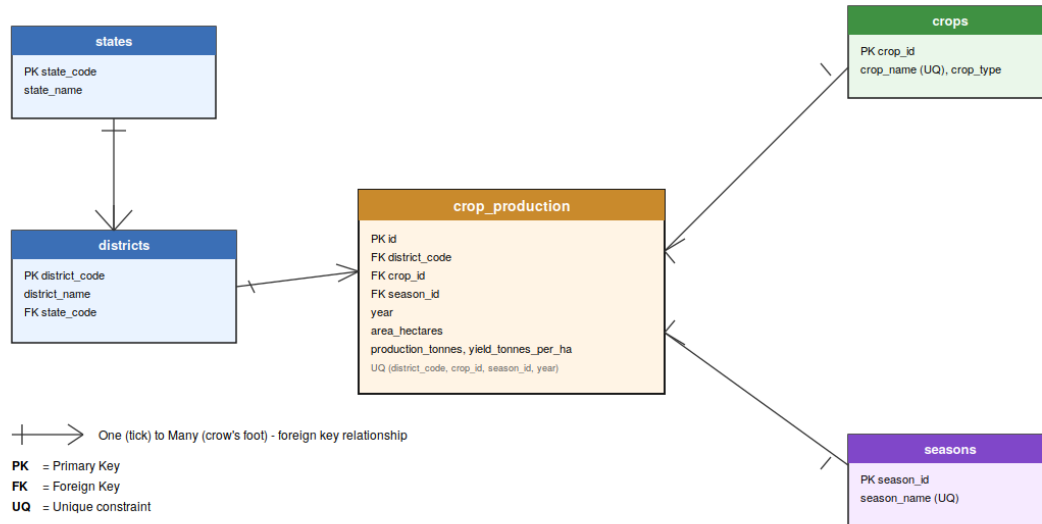
STATE (1) has (N) DISTRICT, and every DISTRICT must belong to exactly one STATE (total participation, shown with a double line). DISTRICT, CROP, and SEASON jointly participate in the ternary relationship RECORDS, whose own attributes (year, area_hectares, production_tonnes, yield_tonnes_per_ha) describe each individual production record. CROP and SEASON both have total participation in RECORDS — every crop and season in the database exists because it appears in at least one production record.

Modeling crop production as a ternary relationship (rather than as just another entity) is the conceptually correct choice here: it genuinely depends on three independent dimensions (district, crop, season) simultaneously, plus the discriminator year. Its identifying key is therefore the composite (district_code, crop_id, season_id, year) — enforced in the physical schema as a UNIQUE constraint even though a surrogate key (id) is used as the primary key for implementation convenience.

D. Logical Design — Relational Schema

The conceptual ER model is mapped to five relational tables: four dimension tables plus one fact table. The diagram below shows this logical design using crow's-foot notation.

Figure 2: Relational Schema Diagram (Logical Design) - Crow's Foot Notation



Every dimension table (states, districts, crops, seasons) connects to the crop_production fact table via a foreign key, enforcing referential integrity - this is the logical-design mapping of the conceptual ER diagram in Figure 1.

Figure 2: Relational Schema Diagram (Crow's Foot Notation)

Data Dictionary

states

Column	Data Type	Constraint	Description
state_code	VARCHAR(10)	PRIMARY KEY	Unique code identifying a state/UT
state_name	VARCHAR(100)	NOT NULL, UNIQUE	Full name of the state/UT

districts

Column	Data Type	Constraint	Description
district_code	VARCHAR(10)	PRIMARY KEY	Unique code identifying a district
district_name	VARCHAR(100)	NOT NULL	Name of the district
state_code	VARCHAR(10)	FOREIGN KEY -> states	Links district to its parent state

crops

Column	Data Type	Constraint	Description
crop_id	INT	PRIMARY KEY, AUTO_INCREMENT	Surrogate key (see Section g for why crop_code was not used)
crop_name	VARCHAR(100)	NOT NULL, UNIQUE	Name of the crop — the true natural identifier
crop_type	VARCHAR(50)	NULLable	Category, e.g. Cereal, Pulses, Oilseeds

seasons

Column	Data Type	Constraint	Description
season_id	INT	PRIMARY KEY, AUTO_INCREMENT	Surrogate key
season_name	VARCHAR(50)	NOT NULL, UNIQUE	e.g. Kharif, Rabi, Whole Year

crop_production (fact table)

Column	Data Type	Constraint	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	Surrogate key for the record
district_code	VARCHAR(10)	FOREIGN KEY -> districts	District where the crop was grown
crop_id	INT	FOREIGN KEY -> crops	Crop grown
season_id	INT	FOREIGN KEY -> seasons	Cropping season
year	VARCHAR(9)	NOT NULL	Agricultural year, e.g. 2015-16
area_hectares	DECIMAL(12,2)	NULLable	Area under cultivation
production_tonnes	DECIMAL(12,2)	NULLable	Total production
yield_tonnes_per_ha	DECIMAL(10,4)	NULLable	Production per unit area

Additional constraint: UNIQUE KEY (district_code, crop_id, season_id, year) — enforces the composite identifying key derived from the conceptual model and prevents duplicate production records for the same combination.

Normalization Justification (1NF -> 3NF)

Stage	Issue in raw data	Fix applied
1NF	Raw CSV has atomic fields but massive repetition (state name repeated per row)	Baseline - already atomic, no repeating groups
2NF	state_name depends only on district, not the full key - a partial dependency	Extracted states and districts into separate tables
3NF	crop_type depends on crop_name, not on the production record - a transitive dependency	Extracted crops and seasons into separate lookup tables

The final crop_production table retains only data that depends on the full key: area, production, and yield.

DDL (Create Table Statements)

```
CREATE TABLE states (  
    state_code VARCHAR(10) PRIMARY KEY,  
    state_name VARCHAR(100) NOT NULL UNIQUE  
);  
  
CREATE TABLE districts (  
    district_code VARCHAR(10) PRIMARY KEY,  
    district_name VARCHAR(100) NOT NULL,  
    state_code VARCHAR(10) NOT NULL,  
    FOREIGN KEY (state_code) REFERENCES states(state_code),  
    UNIQUE KEY uq_district_state (district_name, state_code)  
);  
  
CREATE TABLE crops (  
    crop_id INT AUTO_INCREMENT PRIMARY KEY,  
    crop_name VARCHAR(100) NOT NULL UNIQUE,  
    crop_type VARCHAR(50)  
);  
  
CREATE TABLE seasons (  
    season_id INT AUTO_INCREMENT PRIMARY KEY,  
    season_name VARCHAR(50) NOT NULL UNIQUE  
);  
  
CREATE TABLE crop_production (  
    id INT AUTO_INCREMENT PRIMARY KEY,  
    district_code VARCHAR(10) NOT NULL,  
    crop_id INT NOT NULL,  
    season_id INT NOT NULL,  
    year VARCHAR(9) NOT NULL,  
    area_hectares DECIMAL(12,2),  
    production_tonnes DECIMAL(12,2),  
    yield_tonnes_per_ha DECIMAL(10,4),  
    FOREIGN KEY (district_code) REFERENCES districts(district_code),  
    FOREIGN KEY (crop_id) REFERENCES crops(crop_id),  
    FOREIGN KEY (season_id) REFERENCES seasons(season_id),  
    UNIQUE KEY uq_production_record (district_code, crop_id, season_id, year)  
);
```

E. Implementation — Data Loading

Table Entries Summary (relations $\geq 5 \times N$, records $> 3000 \times N$, $N = 1$)

Table	Row Count	Description
states	34	All Indian states/UTs
districts	740	All districts, linked to their state
crops	~115	Unique crops (after data-quality fix, see Section g)
seasons	7	Kharif, Rabi, Whole Year, etc.
crop_production	~455,338	Fact table - one row per district/crop/season/year combination
Total	~456,234	Well above the required minimum of 3,000

Data source: Area, Production, Yield (APY) dataset, Ministry of Agriculture & Farmers Welfare, Government of India (via data.gov.in / data.desagri.gov.in), covering 1997-2023.

ETL Pipeline Summary

- Read the ~455K-row CSV using pandas
- Standardize column names; replace placeholder values (-, NA, blank) with proper NULLs
- Coerce numeric columns (area, production, yield) with error handling for malformed values
- Deduplicate and load dimension tables first (states, districts, crops, seasons)
- Map crop_name and season_name to their surrogate keys, then batch-insert the fact table in chunks of 5,000 rows using INSERT IGNORE to skip exact duplicates safely

F. Query Development and Examples

Queries are organized from basic to advanced, covering single-table retrieval, multi-table joins with aggregation, and standard subqueries/EXISTS - the range of SQL concepts expected at undergraduate DBMS level.

Section 1: Basic Queries

1.1 List all states

```
SELECT * FROM states ORDER BY state_name;
```

1.2 List all crops of a specific type

```
SELECT crop_name, crop_type FROM crops WHERE crop_type = 'Cereals';
```

1.3 Count total crop production records

```
SELECT COUNT(*) AS total_records FROM crop_production;
```

1.4 All distinct crop types

```
SELECT DISTINCT crop_type FROM crops WHERE crop_type IS NOT NULL;
```

1.5 All production records from a specific year

```
SELECT * FROM crop_production WHERE year = '1997-1998' LIMIT 20;
```

1.6 Number of records per year

```
SELECT year, COUNT(*) AS record_count  
FROM crop_production  
GROUP BY year  
ORDER BY year;
```

1.7 Simple threshold filter

```
SELECT * FROM crop_production  
WHERE production_tonnes > 100000  
ORDER BY production_tonnes DESC  
LIMIT 20;
```

Section 2: Intermediate Queries (Joins, Grouping, HAVING)

2.1 All districts in a given state

```
SELECT d.district_name
FROM districts d
JOIN states s ON d.state_code = s.state_code
WHERE s.state_name = 'Gujarat'
ORDER BY d.district_name;
```

2.2 Total production of a specific crop, all-time

```
SELECT SUM(cp.production_tonnes) AS total_production
FROM crop_production cp
JOIN crops c ON cp.crop_id = c.crop_id
WHERE c.crop_name = 'Rice';
```

2.3 Average area cultivated per crop

```
SELECT c.crop_name, ROUND(AVG(cp.area_hectares), 2) AS avg_area
FROM crop_production cp
JOIN crops c ON cp.crop_id = c.crop_id
GROUP BY c.crop_name
ORDER BY avg_area DESC
LIMIT 10;
```

2.4 Top 10 crop-producing states (all-time)

```
SELECT s.state_name, SUM(cp.production_tonnes) AS total_production
FROM crop_production cp
JOIN districts d ON cp.district_code = d.district_code
JOIN states s ON d.state_code = s.state_code
GROUP BY s.state_name
ORDER BY total_production DESC
LIMIT 10;
```

2.5 Season-wise total production comparison

```
SELECT se.season_name, SUM(cp.production_tonnes) AS total_production
FROM crop_production cp
JOIN seasons se ON cp.season_id = se.season_id
GROUP BY se.season_name
ORDER BY total_production DESC;
```

2.6 Crop diversity per state

```
SELECT s.state_name, COUNT(DISTINCT cp.crop_id) AS distinct_crops_grown
FROM crop_production cp
JOIN districts d ON cp.district_code = d.district_code
JOIN states s ON d.state_code = s.state_code
GROUP BY s.state_name
ORDER BY distinct_crops_grown DESC
LIMIT 10;
```

2.7 Filtering grouped results with HAVING

```
SELECT d.district_name, s.state_name, COUNT(DISTINCT cp.crop_id) AS crop_variety
FROM crop_production cp
JOIN districts d ON cp.district_code = d.district_code
JOIN states s ON d.state_code = s.state_code
GROUP BY d.district_name, s.state_name
HAVING crop_variety > 20
ORDER BY crop_variety DESC
LIMIT 10;
```

2.8 Top 5 districts by average yield for a given crop

```
SELECT d.district_name, s.state_name,  
       ROUND(AVG(cp.yield_tonnes_per_ha), 2) AS avg_yield  
FROM crop_production cp  
JOIN districts d ON cp.district_code = d.district_code  
JOIN states s ON d.state_code = s.state_code  
JOIN crops c ON cp.crop_id = c.crop_id  
WHERE c.crop_name = 'Wheat'  
GROUP BY d.district_name, s.state_name  
HAVING COUNT(*) > 5  
ORDER BY avg_yield DESC  
LIMIT 5;
```

2.9 CASE WHEN - bucket production into size categories

```
SELECT cp.id, cp.production_tonnes,  
       CASE  
         WHEN cp.production_tonnes >= 100000 THEN 'Very High'  
         WHEN cp.production_tonnes >= 10000  THEN 'High'  
         WHEN cp.production_tonnes >= 1000   THEN 'Medium'  
         ELSE 'Low'  
       END AS production_category  
FROM crop_production cp  
LIMIT 20;
```

2.10 Year-over-year production trend for a specific crop

```
SELECT cp.year, SUM(cp.production_tonnes) AS total_production  
FROM crop_production cp  
JOIN crops c ON cp.crop_id = c.crop_id  
WHERE c.crop_name = 'Rice'  
GROUP BY cp.year  
ORDER BY cp.year;
```

Section 3: Advanced Queries (Subqueries, EXISTS)

3.1 Crops whose average yield beats the overall average yield

```
SELECT c.crop_name,  
       ROUND(AVG(cp.yield_tonnes_per_ha), 2) AS crop_avg_yield  
FROM crop_production cp  
JOIN crops c ON cp.crop_id = c.crop_id  
GROUP BY c.crop_name  
HAVING crop_avg_yield > (SELECT AVG(yield_tonnes_per_ha) FROM crop_production)  
ORDER BY crop_avg_yield DESC  
LIMIT 15;
```

3.2 States that have ever produced more than 1,000,000 tonnes of any single crop (EXISTS)

```
SELECT s.state_name  
FROM states s  
WHERE EXISTS (  
    SELECT 1  
    FROM crop_production cp  
    JOIN districts d ON cp.district_code = d.district_code  
    WHERE d.state_code = s.state_code  
    AND cp.production_tonnes > 1000000  
)  
ORDER BY s.state_name;
```

3.3 Crops grown in every single season

```
SELECT c.crop_name  
FROM crop_production cp  
JOIN crops c ON cp.crop_id = c.crop_id  
GROUP BY c.crop_name  
HAVING COUNT(DISTINCT cp.season_id) = (SELECT COUNT(*) FROM seasons);
```

3.4 District(s) with the single highest production value recorded

```
SELECT d.district_name, s.state_name, c.crop_name, cp.year, cp.production_tonnes  
FROM crop_production cp  
JOIN districts d ON cp.district_code = d.district_code  
JOIN states s ON d.state_code = s.state_code  
JOIN crops c ON cp.crop_id = c.crop_id  
WHERE cp.production_tonnes = (SELECT MAX(production_tonnes) FROM crop_production);
```

Results:

state_code	state_name
35	Andaman And Nicobar Islands
28	Andhra Pradesh
12	Arunachal Pradesh
18	Assam
10	Bihar
4	Chandigarh
22	Chhattisgarh
30	Goa
24	Gujarat
6	Haryana
2	Himachal Pradesh
1	Jammu And Kashmir
20	Jharkhand
29	Karnataka
32	Kerala
37	Ladakh
23	Madhya Pradesh
27	Maharashtra
14	Manipur
17	Meghalaya
15	Mizoram
13	Nagaland
21	Odisha
34	Puducherry
3	Punjab
8	Rajasthan
11	Sikkim
33	Tamil Nadu
36	Telangana
38	The Dadra And Nagar Haveli...
16	Tripura
9	Uttar Pradesh
5	Uttarakhand
19	West Bengal
NULL	NULL

1.1 List all States

crop_name	crop_type
Bajra	Cereals
Jowar	Cereals
Korra	Cereals
Malze	Cereals
Ragi	Cereals
Rice	Cereals
Wheat	Cereals
Samai	Cereals
Small Millets	Cereals
Varagu	Cereals
Barley	Cereals
Other Cere...	Cereals
Jobster	Cereals

1.2 List all crops of a specific type

total_records
455359

1.3 Count total crop production records

crop_type
Pulses
Cereals
Oilseeds
Fiber Crops
Spices
Sugar
Drugs And Narcotics
Vegetable
Fruits
Plantation Crops

1.4 All distinct crop types

id	district_code	crop_id	season_id	year	area_hectares	production_tonn...	yield_tonnes_per_...	
1	502	1	1	1997-1998	21400.00	2600.00	0.1210	
2	502	2	1	1997-1998	1400.00	500.00	0.3570	
3	502	3	1	1997-1998	1000.00	100.00	0.1000	
4	502	4	1	1997-1998	7300.00	9400.00	1.2880	
5	502	5	1	1997-1998	3700.00	7100.00	1.9190	
6	502	5	2	1997-1998	100.00	100.00	1.0000	
7	502	5	3	1997-1998	3800.00	7200.00	1.8950	
8	502	6	2	1997-1998	28000.00	14800.00	0.5290	
9	502	7	1	1997-1998	650800.00	228400.00	0.3510	
10	502	7	2	1997-1998	20200.00	21700.00	1.0740	
11	502	7	3	1997-1998	671000.00	250100.00	0.3730	
12	502	8	1	1997-1998	3300.00	1000.00	0.3030	
13	502	8	2	1997-1998	600.00	200.00	0.3330	
14	502	8	3	1997-1998	3900.00	1200.00	0.3080	
15	502	9	1	1997-1998	10100.00	10200.00	1.0100	
16	502	9	2	1997-1998	18800.00	9400.00	0.5000	
17	502	9	3	1997-1998	28900.00	19600.00	0.6780	
18	502	10	1	1997-1998	2200.00	700.00	0.3180	
19	502	10	2	1997-1998	100.00	100.00	1.0000	
20	502	10	3	1997-1998	2300.00	800.00	0.3480	

1.5 All production records from a specific year

year	record_count	
1997-1998	10445	
1998-1999	13670	
1999-2000	14329	
2000-2001	14813	
2001-2002	14406	
2002-2003	17795	
2003-2004	18436	
2004-2005	15350	
2005-2006	15239	
2006-2007	16073	
2007-2008	16094	
2008-2009	16641	
2009-2010	16975	
2010-2011	16531	
2011-2012	18128	
2012-2013	17227	
2013-2014	18618	
2014-2015	18022	
2015-2016	18473	
2016-2017	19801	
2017-2018	20556	
2018-2019	20744	
2019-2020	22186	
2020-2021	21608	
2021-2022	21350	
2022-2023	21849	

1.6 Number of records per year

id	district_code	crop_id	season_id	year	area_hectares	production_tonnes	yield_tonnes_per_ha	
162394	569	29	4	2007-2008	107106.00	1597800000.00	14917.9320	
442433	548	29	4	2022-2023	220017.00	1583751764.00	7198.3150	
420768	548	29	4	2021-2022	170889.00	1498701000.00	8770.0260	
383829	569	29	4	2019-2020	87749.00	1488200000.00	16959.7370	
146332	569	29	4	2006-2007	104197.00	1482900000.00	14231.6960	
377033	548	29	4	2019-2020	176243.00	1452725000.00	8242.7390	
448374	569	29	4	2022-2023	91809.00	1425200000.00	15523.5330	
301886	569	29	4	2015-2016	85448.00	1401300000.00	16399.4480	
242249	548	29	4	2012-2013	147539.00	1393593000.00	9445.5910	
224267	548	29	4	2011-2012	143110.00	1351758000.00	9445.5870	
207169	548	29	4	2010-2011	142248.00	1347700000.00	9474.2980	
405037	569	29	4	2020-2021	88467.00	1323700000.00	14962.6410	
355908	548	29	4	2018-2019	183232.00	1305171000.00	7123.0520	
278440	548	29	4	2014-2015	149419.00	1283682000.00	8591.2230	
296338	548	29	4	2015-2016	152341.00	1264153000.00	8298.1800	
230074	569	29	4	2011-2012	82704.00	1250800000.00	15123.8150	
361967	569	29	4	2018-2019	87702.00	1250600000.00	14259.6520	
178958	569	29	4	2008-2009	76902.00	1240600000.00	16132.2200	
426319	569	29	4	2021-2022	89927.00	1219800000.00	13564.3360	
265511	569	29	4	2013-2014	84531.00	1212000000.00	14337.9350	
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	

1.7 Simple filter with a threshold

district_name	
Ahmedabad	
Amreli	
Anand	
Arvali	
Banas Kantha	
Bharuch	
Bhavnagar	
Botad	
Chhotaudepur	
Dahod	
Dangs	
Devbhumi Dwarka	
Gandhinagar	
Gir Somnath	
Jamnagar	
Junagadh	
Kachchh	
Kheda	
Mahesana	
Mahisagar	
Morbi	
Narmada	
Navsari	
Panch Mahals	
Patan	
Porbandar	
Rajkot	
Sabar Kantha	
Surat	
Surendranagar	
Tapi	
Vadodara	
Valest	

2.1 All districts in a given state

total_production	
4258304548.90	

2.2 Total production of a specific crop, all-time

crop_name	avg_area	
Wheat	59600.05	
Rice	57705.68	
Soyabean	41057.77	
Cotton(Lint)	34703.03	
Rubber	32797.79	
Guar Seed	32042.63	
Coffee	27984.67	
Jowar	24456.15	
Bajra	23636.35	
Arcanut (Processed)	19816.26	

2.3 Average area cultivated per crop

state_name	total_production	
Kerala	140713787117.52	
Tamil Nadu	95887012719.25	
Karnataka	75236372475.48	
Andhra Pradesh	31826598702.64	
West Bengal	9502153095.70	
Uttar Pradesh	5578466113.73	
Assam	4273490101.00	
Maharashtra	2578003575.02	
Andaman And Nicobar Islands	2216332338.86	
Goa	1787683048.50	

2.4 Top 10 crop-producing states (all-time)

season_name	total_production	
Whole Year	362907093062.51	
Kharif	6978795524.56	
Rabi	3836628719.59	
Total	2586802297.02	
Winter	702136616.79	
Summer	297339123.96	
Autumn	98908629.67	

2.5 Season-wise total production comparison

state_name	distinct_crops_grown	
Tamil Nadu	86	
Andhra Pradesh	68	
Telangana	62	
Karnataka	61	
Madhya Pradesh	61	
Kerala	54	
Rajasthan	50	
Manipur	49	
Chhattisgarh	47	
Nagaland	45	

2.6 Crop diversity per state

district_name	state_name	crop_variety	
Dindigul	Tamil Nadu	81	
Salem	Tamil Nadu	79	
Erode	Tamil Nadu	75	
Vellore	Tamil Nadu	75	
Coimbatore	Tamil Nadu	74	
Dharmapuri	Tamil Nadu	74	
Namakkal	Tamil Nadu	74	
Madurai	Tamil Nadu	72	
Theni	Tamil Nadu	72	
Krishnagiri	Tamil Nadu	71	

2.7 Filtering grouped results with HAVING

district_name	state_name	avg_yield	
Barnala	Punjab	5.02	
Sangrur	Punjab	4.88	
Moga	Punjab	4.87	
Ludhiana	Punjab	4.81	
Fatehgarh Sahib	Punjab	4.73	

2.8 Top 5 districts by average yield for a given crop

id	production_tonnes	production_cat...	
1	2600.00	Medium	
2	500.00	Low	
3	100.00	Low	
4	9400.00	Medium	
5	7100.00	Medium	
6	100.00	Low	
7	7200.00	Medium	
8	14800.00	High	
9	228400.00	Very High	
10	21700.00	High	
11	250100.00	Very High	
12	1000.00	Medium	
13	200.00	Low	
14	1200.00	Medium	
15	10200.00	High	
16	9400.00	Medium	
17	19600.00	High	
18	700.00	Low	
19	100.00	Low	
20	800.00	Low	

2.9 CASE WHEN — bucket production into size categories

year	total_production	
1997-1998	111444707.00	
1998-1999	134705996.00	
1999-2000	131407796.60	
2000-2001	134479491.80	
2001-2002	136514624.00	
2002-2003	111596357.00	
2003-2004	134392720.75	
2004-2005	129546992.71	
2005-2006	138673817.29	
2006-2007	146583109.32	
2007-2008	151591129.33	
2008-2009	156356679.59	
2009-2010	141664616.49	
2010-2011	153013749.13	
2011-2012	166857198.16	
2012-2013	170938539.01	
2013-2014	175731921.36	
2014-2015	180215080.68	
2015-2016	157930935.90	
2016-2017	187234032.66	
2017-2018	188539114.20	
2018-2019	187984357.68	
2019-2020	221555453.81	
2020-2021	233609352.99	
2021-2022	222441319.50	
2022-2023	253295455.94	

2.10 Year-over-year production trend for a specific crop

crop_name	crop_avg_yield	
Coconut	8876.88	

3.1 Crops whose average yield beats the overall average yield

state_name	
Andaman And Nicobar Islands	
Andhra Pradesh	
Assam	
Bihar	
Chhattisgarh	
Goa	
Gujarat	
Haryana	
Karnataka	
Kerala	
Madhya Pradesh	
Maharashtra	
Odisha	
Puducherry	
Punjab	
Rajasthan	
Tamil Nadu	
Telangana	
Uttar Pradesh	
Uttarakhand	
West Bengal	

3.2 States that have ever produced more than 1,000,000 tonnes of any single crop (EXISTS)

crop_name	
Arhar/Tur	
Banana	
Cotton(Lint)	
Dry Chillies	
Ginger	
Groundnut	
Maize	
Moong(Green Gram)	
Onion	
Peas & Beans (Pulses)	
Potato	
Ragi	
Rice	
Sesamum	
Soyabean	
Sugarcane	
Turmeric	
Urad	

3.3 Crops grown in every single season (subquery with COUNT comparison)

district_name	state_name	crop_name	year	production_tonnes	
Coimbatore	Tamil Nadu	Coconut	2007-2008	1597800000.00	

3.4 District(s) with the single highest production value recorded

G. Testing and Evaluation

1. Data Integrity Testing — the crop_code discovery

During loading, crops.crop_code (the field originally assumed to be the natural key) was found to be non-unique in the raw government dataset — a single code mapped to as many as 7 different crop names. This was discovered by comparing `df['crop_code'].nunique()` (94) against `df['crop_name'].nunique()` (115) in the raw data. The schema was corrected to use a surrogate key (crop_id) with crop_name as the true unique field, and the ETL pipeline was rewritten to re-map records accordingly.

2. Referential Integrity Testing

Verified that foreign key constraints reject invalid references:

```
-- Attempting to insert a production record with a non-existent crop_id
-- should FAIL with a foreign key constraint error:
INSERT INTO crop_production (district_code, crop_id, season_id, year, production_tonnes)
VALUES ('001', 999999, 1, '2023-24', 5000);
-- Expected: Error 1452 (23000): Cannot add or update a child row:
-- a foreign key constraint fails
```

3. Constraint Testing (Uniqueness)

Verified that the UNIQUE KEY on crop_production prevents duplicate records for the same district/crop/season/year:

```
-- Re-running the loader a second time uses INSERT IGNORE, so duplicate
-- rows matching the existing UNIQUE KEY are silently skipped rather
-- than causing duplicate data. Verified: re-running load_data.py twice
-- produced the same final row count both times.
```

4. Load Verification

```
SELECT COUNT(*) FROM states;           -- 34
SELECT COUNT(*) FROM districts;        -- 740
SELECT COUNT(*) FROM crops;            -- ~115
SELECT COUNT(*) FROM seasons;         -- 7
SELECT COUNT(*) FROM crop_production;  -- ~455,338
```

5. Functional Testing

All 21 queries (Section f) were executed against the loaded dataset and verified to return semantically correct, non-empty results with properly joined state/district/crop/season names.

Challenges Faced and Resolutions

Challenge	Resolution
mysql-connector-python wrote the literal word 'nan' into INSERT statements for missing numeric values	Explicitly converted NaN -> None before insertion so MySQL correctly interprets them as NULL
crop_code was assumed to be a natural key but was found to be non-unique	Redesigned crops table around a surrogate key (crop_id) with crop_name as the true unique identifier
Loading ~455K rows in a single transaction risked memory issues and made progress hard to track	Batched inserts (5,000 rows per batch) with progress logging

H. Conclusion

This project successfully models and populates a normalized relational database for Indian agricultural crop production data, sourced from an authoritative government dataset spanning 27 years and 740 districts. The design demonstrates justified normalization (1NF -> 3NF) grounded in the actual structure of the data, a conceptually correct ternary-relationship model for the fact table, referential integrity via foreign keys and uniqueness constraints, a working automated ETL pipeline handling real-world data quality issues, and a graded progression of SQL queries from basic retrieval to subquery-based analysis.

Unique Features (compared to typical fresher DBMS projects)

- Uses a real, large-scale, authoritative government dataset (not synthetic/Kaggle-toy data), spanning nearly three decades
- Surfaced and resolved a genuine data integrity defect in the source data (non-unique crop_code), rather than assuming the raw data was clean
- Implements an automated Python ETL pipeline with proper NULL-handling and batch processing for ~455K records, rather than manual data entry

Future Work

- Extend the schema with rainfall/weather data to analyze climate impact on yield
- Expose the database via a simple web dashboard for interactive querying
- Add indexing and query optimization as a follow-up performance-tuning phase